

NYCU-ECE DCS-2026

HW04

Design: Multi-Head Attention

Data Preparation

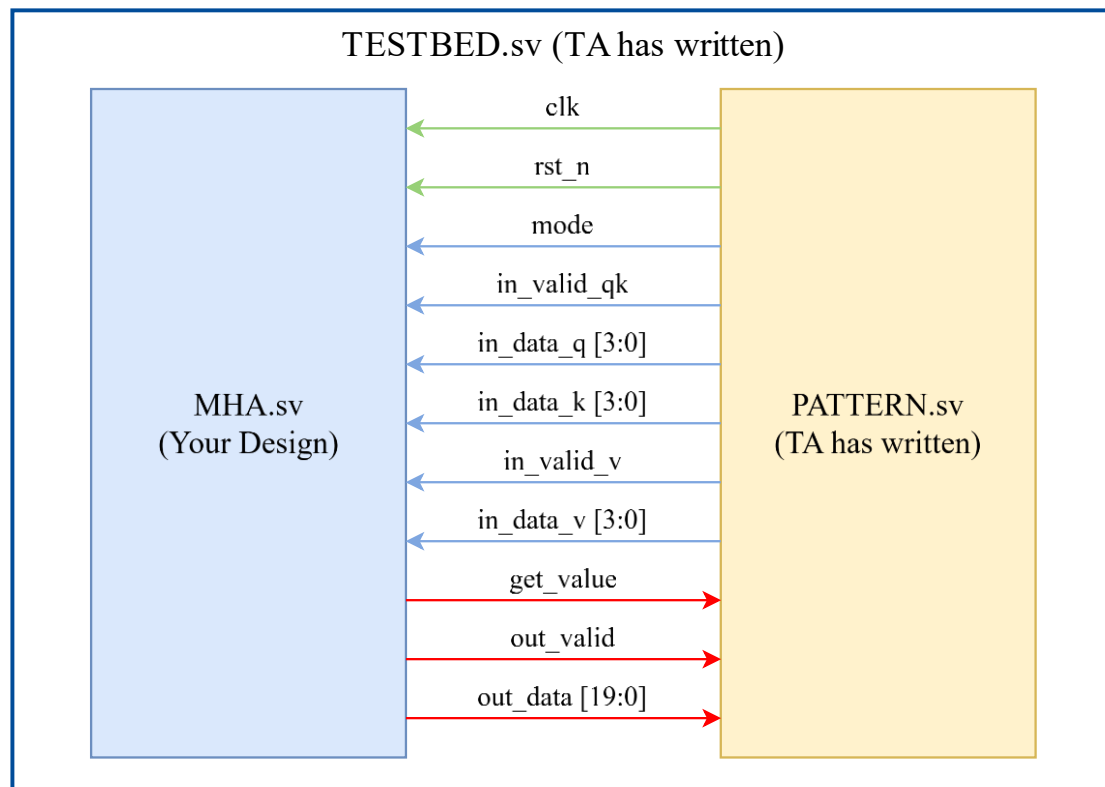
1. Extract the files needed from TA's directory:

```
$ tar -xvf ~es26dcsTA01/HW04.tar
```

2. The extracted files contain:

```
HW04/  
├── 00_TESTBED/  
├── 01_RTL/  
├── 02_SYN/  
├── 03_GATE/  
└── 09_UPLOAD/
```

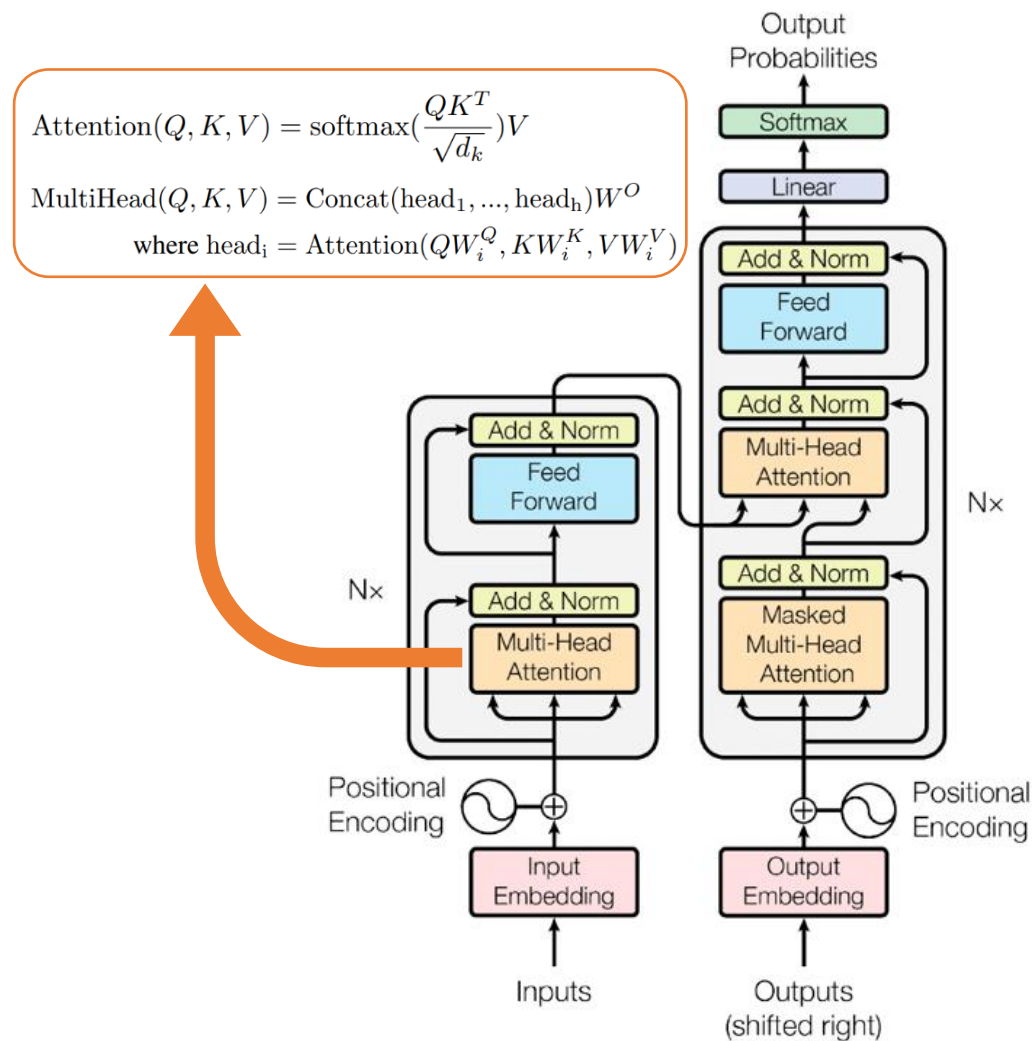
Block Diagram



MHA Introduction

In **LLM (Large Language Model)**, self-attention is a very important part, because it enables each token to dynamically “attend” to other tokens in the sequence and aggregate contextual information through weighted interactions. Compared with a single-head attention mechanism, **Multi-Head Attention (MHA)** computes attention in multiple subspaces in parallel by splitting the feature dimension into several heads, allowing the model to capture diverse relationships (e.g., syntax, long-range dependency, and semantic similarity) simultaneously.

In hardware design, MHA can be mapped as two main matrix-multiplication stages — $\mathbf{Q} \cdot \mathbf{K}^T$, where $\mathbf{Q}$ means Queries and $\mathbf{K}$ means Keys, followed by an activation/normalization step and then $\mathbf{S} \cdot \mathbf{V}$, where $\mathbf{S}$ means Softmax results and $\mathbf{V}$ means Values — and the head-wise partitioning provides flexibility to trade off area and latency via parallel execution or resource sharing.



Reference: Attention Is All You Need, <https://arxiv.org/abs/1706.03762>

Design Description

In this HW, you need to implement a Multi-Head Attention (MHA) accelerator that performs the core attention computation in a streaming and hardware-friendly manner, including two matrix multiplications: $\mathbf{S} = \mathbf{Q} \times \mathbf{K}^T$, where $\mathbf{K}^T$ is the Transpose of $\mathbf{K}$ and $\mathbf{O} = \text{act}(\mathbf{S}) \times \mathbf{V}$, where $\text{act}(\cdot)$ is an activation stage (Softmax is not required in this HW).

The design must support two modes: **(1)** single-head mode, computing one full 8×8 attention and producing an 8×8 output; and **(2)** two-head mode, partitioning the feature dimension into two heads (each head has width 4), computing two head outputs $[8 \times 4]$, and then concatenating them to form the final 8×8 output.

● single-head mode (mode = 0)

1. You will receive $\mathbf{Q} [8 \times 8]$ and $\mathbf{K} [8 \times 8]$, then you have to compute

$$\mathbf{S} [8 \times 8] = \mathbf{Q} [8 \times 8] \times \mathbf{K}^T [8 \times 8]$$

Q								K								S = Q * K ^T							
-2	-7	5	-2	3	-7	-8	-1	-3	-4	-2	-6	3	-2	3	3	32	25	147	-103	-47	-23	-106	-50
6	-8	4	7	-1	4	-3	6	6	-6	3	2	3	-3	6	-2	-38	65	-57	66	-9	-69	-48	-115
3	-5	-8	0	4	2	-7	-2	-1	-8	-5	-6	6	-8	-3	-4	8	-8	114	-21	13	-9	22	10
7	-6	0	7	-4	3	-3	-6	4	0	-4	7	-6	0	5	3	-84	65	-16	68	-32	-86	-17	3
2	1	0	7	-4	-5	-1	1	1	6	-5	7	4	-4	0	4	-54	15	-37	79	65	-18	6	-68
4	-3	5	0	5	5	2	-8	4	1	-4	-6	3	-6	6	3	-23	85	11	-48	-71	-34	-89	32
3	2	-2	3	4	4	1	3	5	5	-8	-5	-7	1	0	3	-15	6	-50	31	58	7	11	-56
-6	7	-3	6	-1	-1	-1	-2	-4	-3	-5	-7	-5	3	5	-8	-50	-77	-58	25	85	-50	-1	-11

2. Apply the activation function $\text{act}(\cdot)$ element-wise to $\mathbf{S}$, then you get $\mathbf{P}$,

$$\text{where } \text{act}(x) = \begin{cases} x, & x \geq 0 \\ x/4 \text{ (truncate toward zero)}, & \text{otherwise} \end{cases}$$

S = Q * K ^T								P = act(S)							
32	25	147	-103	-47	-23	-106	-50	32	25	147	-25	-11	-5	-26	-12
-38	65	-57	66	-9	-69	-48	-115	-9	65	-14	66	-2	-17	-12	-28
8	-8	114	-21	13	-9	22	10	8	-2	114	-5	13	-2	22	10
-84	65	-16	68	-32	-86	-17	3	-21	65	-4	68	-8	-21	-4	3
-54	15	-37	79	65	-18	6	-68	-13	15	-9	79	65	-4	6	-17
-23	85	11	-48	-71	-34	-89	32	-5	85	11	-12	-17	-8	-22	32
-15	6	-50	31	58	7	11	-56	-3	6	-12	31	58	7	11	-14
-50	-77	-58	25	85	-50	-1	-11	-12	-19	-14	25	85	-12	0	-2

3. Using `get_value` to get $\mathbf{V} [8 \times 8]$.

$$\text{Then compute the output matrix: } \text{out_data} [8 \times 8] = \mathbf{O} [8 \times 8] = \mathbf{P} [8 \times 8] \times \mathbf{V} [8 \times 8]$$

P = act(S)								V								out_data = P * V							
32	25	147	-25	-11	-5	-26	-12	5	-7	-6	1	-5	-7	-3	0	-93	-642	-554	-748	186	-524	587	70
-9	65	-14	66	-2	-17	-12	-28	0	4	-6	7	2	-6	5	3	279	736	145	353	448	309	-285	465
8	-2	114	-5	13	-2	22	10	-2	-4	0	-8	1	0	2	1	-173	-623	47	-1008	-214	-38	135	174
-21	65	-4	68	-8	-21	-4	3	0	2	5	-5	4	6	-8	4	-27	628	69	12	478	323	-161	644
-13	15	-9	79	65	-4	6	-17	3	-3	4	-1	-6	-4	-7	-2	320	264	787	-203	-56	385	-1043	221
-5	85	11	-12	-17	-8	-22	32	-6	-3	-3	5	5	-8	3	-8	-350	190	-886	337	396	-739	739	346
-3	6	-12	31	58	7	11	-14	2	0	5	-1	-8	6	-1	3	275	44	487	44	-262	33	-657	-51
-12	-19	-14	25	85	-12	0	-2	-8	-6	-6	-7	0	-6	2	3	311	-93	699	-289	-462	116	-922	-51

● **two-head mode (mode = 1)**

1. You will receive Q [8×8] and K [8×8]. In two-head mode, you must split Q and K along the feature (column) dimension into Q₁/Q₂ and K₁/K₂, where each sub-matrix is [8×4]. Then compute the attention score matrices for each head:

$$S_1[8 \times 8] = Q_1[8 \times 4] \times K_1^T[4 \times 8]; S_2[8 \times 8] = Q_2[8 \times 4] \times K_2^T[4 \times 8]$$

Q1				Q2				S1 = Q1 * K1T							
-2	-7	5	-2	3	-7	-8	-1	36	41	45	-42	-83	-23	-75	18
6	-8	4	7	-1	4	-3	6	-36	110	-4	57	-13	-42	-77	-69
3	-5	-8	0	4	2	-7	-2	27	24	77	44	13	39	54	43
7	-6	0	7	-4	3	-3	-6	-39	92	-1	77	20	-20	-30	-59
2	1	0	7	-4	-5	-1	1	-52	20	-52	57	57	-33	-20	-60
4	-3	5	0	5	5	2	-8	-10	57	-5	-4	-39	-7	-35	-32
3	2	-2	3	4	4	1	3	-31	6	-27	41	46	4	26	-29
-6	7	-3	6	-1	-1	-1	-2	-40	-75	-71	30	93	-41	-1	-24
K1				K2				S2 = Q2 * K2T							
-3	-4	-2	-6	3	-2	3	3	-4	-16	102	-61	36	0	-31	-68
6	-6	3	2	3	-3	6	-2	-2	-45	-53	9	4	-27	29	-46
-1	-8	-5	-6	6	-8	-3	-4	-19	-32	37	-65	0	-48	-32	-33
4	0	-4	7	-6	0	5	3	-45	-27	-15	-9	-52	-66	13	62
1	6	-5	7	4	-4	0	4	-2	-5	15	22	8	15	26	-8
4	1	-4	-6	3	-6	6	3	-13	28	16	-44	-32	-27	-54	64
5	5	-8	-5	-7	1	0	3	16	0	-23	-10	12	3	-15	-27
-4	-3	-5	-7	-5	3	5	-8	-10	-2	13	-5	-8	-9	0	13

2. Apply the activation function **act(·)** element-wise to **S₁** and **S₂**, then you get **P₁**

and **P₂**, where
$$act(x) = \begin{cases} x, & x \geq 0 \\ x/4 \text{ (truncate toward zero)}, & \text{otherwise} \end{cases}$$

S1 = Q1 * K1T								P1 = act (S1)							
36	41	45	-42	-83	-23	-75	18	36	41	45	-10	-20	-5	-18	18
-36	110	-4	57	-13	-42	-77	-69	-9	110	-1	57	-3	-10	-19	-17
27	24	77	44	13	39	54	43	27	24	77	44	13	39	54	43
-39	92	-1	77	20	-20	-30	-59	-9	92	0	77	20	-5	-7	-14
-52	20	-52	57	57	-33	-20	-60	-13	20	-13	57	57	-8	-5	-15
-10	57	-5	-4	-39	-7	-35	-32	-2	57	-1	-1	-9	-1	-8	-8
-31	6	-27	41	46	4	26	-29	-7	6	-6	41	46	4	26	-7
-40	-75	-71	30	93	-41	-1	-24	-10	-18	-17	30	93	-10	0	-6
S2 = Q2 * K2T								P2 = act (S2)							
-4	-16	102	-61	36	0	-31	-68	-1	-4	102	-15	36	0	-7	-17
-2	-45	-53	9	4	-27	29	-46	0	-11	-13	9	4	-6	29	-11
-19	-32	37	-65	0	-48	-32	-33	-4	-8	37	-16	0	-12	-8	-8
-45	-27	-15	-9	-52	-66	13	62	-11	-6	-3	-2	-13	-16	13	62
-2	-5	15	22	8	15	26	-8	0	-1	15	22	8	15	26	-2
-13	28	16	-44	-32	-27	-54	64	-3	28	16	-11	-8	-6	-13	64
16	0	-23	-10	12	3	-15	-27	16	0	-5	-2	12	3	-3	-6
-10	-2	13	-5	-8	-9	0	13	-2	0	13	-1	-2	-2	0	13

Specifications

1. Top module name: **MHA** (File name: **MHA.sv**).
2. After asynchronous active-low reset, all output signals must be reset to zero.
3. The 02_SYN result **must not have any errors** and **must not contain any latches**.
4. The slack in the timing report must be non-negative, and the result must be **"Met"**.
5. Gate level simulation **must not have any timing violation**.
6. The **clock cycle time** can be adjusted manually, with a maximum precision of **one decimal place**. You must clearly state your chosen cycle time when submitting the assignment. For testing purposes, you are required to modify the cycle time specified in the following locations:

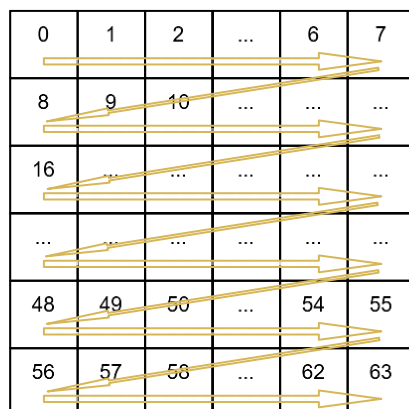
``define CYCLE_TIME 10.0` (00_TESTBED / *PATTERN.sv*: line 1)

`set CYCLE 10.0` (02_SYN / *syn.tcl*: line 11)

7. Input delay = $0.5 \times \text{clock period}$, output delay = $0.5 \times \text{clock period}$.
8. Do not use `xerrorx`, `xlatchx`, `xcongratulationx` or `xfailx` as names for logic / wire / reg / submodule / parameter, otherwise the demo result will fail.
(Note: `x` represents any symbol before or after the word. Ex: `error_test` is prohibited.)
9. **You can use for loops** in this homework.
10. The demonstration will use **a hidden test pattern**. Ensure your design can handle various random traffic conditions, not just the test cases provided.

Important Notice (Look Carefully)

1. *in_data_q*, *in_data_k*, and *in_data_v* are all sent in raster-scan order for their 8×8 matrices; *out_data* is also output in raster-scan order for the 8×8 result matrix.



(raster-scan order)

2. *out_data* must be zero when *out_valid* is low.
3. In practice, the following constraints are also fundamentally required, because the

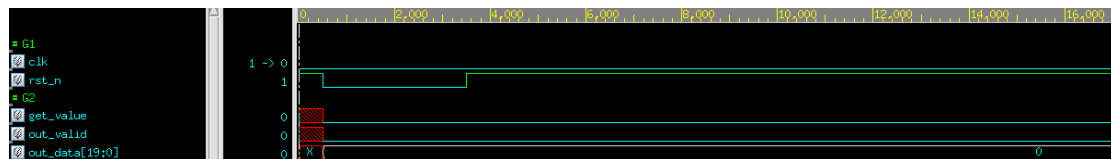
DUT must have received the required V data before producing correct outputs.

- No overlap between `get_value` and `out_valid`.
- `out_valid` must **NOT** assert before `get_value`.

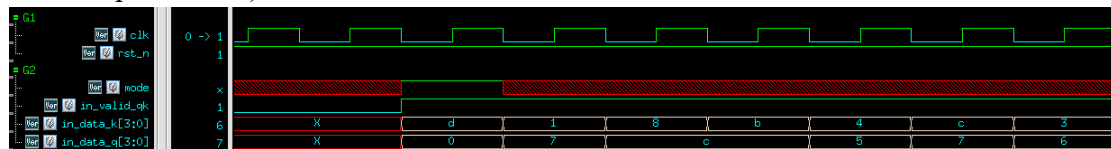
4. `get_value` must be a **one-cycle pulse** and occur **only once** per pattern
5. Latency for one single pattern is defined as the duration from [the rising edge of `in_valid_qk`] to [the falling edge of `out_valid`].

Example waveform

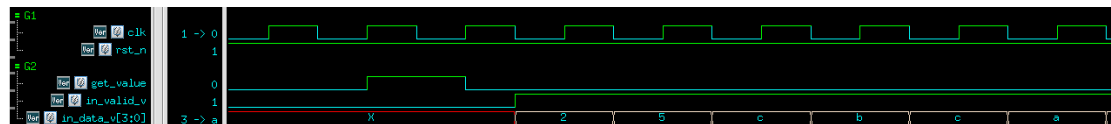
Reset all output after asynchronous and active-low reset signal.



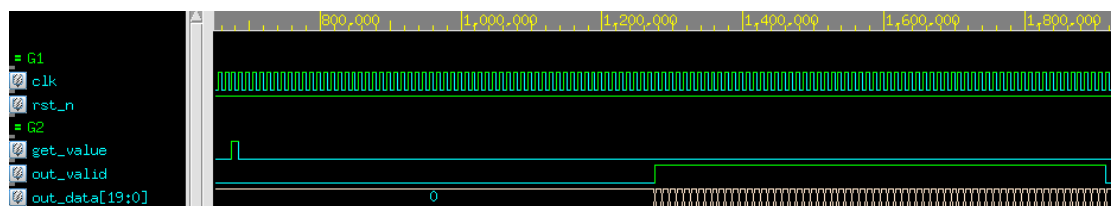
`mode` is only valid in the first cycle when `in_valid_qk` is asserted (i.e., the first cycle of the input stream).



After `get_value` is asserted, `in_valid_v` and `in_data_v` will be asserted **one cycle later** and will remain valid for **64 consecutive cycles**.



`out_valid` and `out_data` must be asserted after `get_value`, and `out_valid` should remain high continuously for 64 consecutive cycles



Upload file

- Use **./01_upload** under 09_UPLOAD/ to submit your design & Submit your report to E3 (named `hw4_report_dcsXXX.pdf`).
- Note that the script requires time to synchronize with the NTP server. Please avoid last-minute submissions.
- 1de deadline: 2026/04/30 23:59:59; 2de deadline: 2026/05/05 15:30:00.
- **Incorrect filenames will result in a 5-point deduction.**

Grading policy

1. Pass the RTL& Synthesis simulation: 60%
2. Performance: 30%
Ranking formula: $\text{execution cycles} \times \text{clock period} \times (\text{area})^2$ (less is better)
3. Report: 10%
4. For submission in 2nd demo, total score will get **40% off**.

Note

Template folders and reference commands:

1. 01_RTL/ (RTL simulation) → **./01_run_vcs_rtl**
2. 02_SYN/ (synthesis) → **./01_run_dc**
3. 03_GATE/ (gate-level simulation) → **./01_run_vcs_gate**
4. 09_UPLOAD/ (upload) → **./01_upload**